Lecture 7: Molecules of life

Escherichia coli (E. coli) is a Gram-negative bacterium that lives in the gut of Homo sapiens.

Photosynthesis is summarised by the equation $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$, which takes place in the chloroplast.

Each water molecule, H_2O, has a bond angle of about 104.5°. Glucose, $C_6H_{12}O_6$, has a molar mass of 180.16 g/mol.

The sulfate ion SO_4^{2-} carries a charge of -2, while the calcium ion Ca^{2+} carries a charge of $+2$.

Carbonic acid dissociates reversibly: $H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$.

The tumour suppressor gene BRCA1 is linked to breast cancer, and the protein p53 halts the cell cycle when DNA is damaged.

The Na^+/K^+-ATPase pumps 3 Na^+ ions out of the cell and 2 K^+ ions in for every ATP hydrolysed.

Lecture 7: Energy and structure

Gibbs free energy change is given by $\Delta G = \Delta H - T\Delta S$; a reaction is spontaneous when ΔG is negative.

Einstein's mass–energy equivalence is written $E = mc^2$.

The acidity of a solution is defined as $\mathrm{pH} = -\log[\mathrm{H^+}]$; human arterial blood has a pH of 7.35–7.45.

In proteins, the α-helix has 3.6 residues per turn, and β-sheets are held together by hydrogen bonds between strands.

A typical animal cell is 10–30 µm across, and the resting membrane potential is −70 mV.

Enzymes in the human body work best at 37 °C; the concentration of glucose in blood is about 5.5 mmol·L⁻¹.

The rate constant k doubled from 0.012 $\mathrm{s^{-1}}$ to 0.024 $\mathrm{s^{-1}}$ when the temperature rose by 10 °C.

Lecture 7: Word-style notation

Respiration releases energy: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$.

Calcium ions (Ca^{2+}) trigger muscle contraction; the speed of light c is 3.0×10^{8} m/s.

Mitochondrial efficiency is limited because some energy is lost as heat during oxidative phosphorylation.

Water freezes at 0 °C and the lowest temperature on the Kelvin scale is −273.15 °C.